

Fixed-Time Cooperative Tracking Control for Double-Integrator Multiagent Systems: A Time-Based Generator Approach

Qiang Chen[✉], Yu Zhao[✉], *Member, IEEE*, Guanghui Wen[✉], *Senior Member, IEEE*, Guoqing Shi[✉],
and Xinghuo Yu[✉], *Fellow, IEEE*

Abstract—In this article, both the fixed-time distributed consensus tracking and the fixed-time distributed average tracking problems for double-integrator-type multiagent systems with bounded input disturbances are studied. First, a new practical robust fixed-time sliding-mode control method based on the time-based generator is proposed. Second, two fixed-time distributed consensus tracking observers for double-integrator-type multiagent systems are designed to estimate the state disagreement between the leader and the followers under undirected and directed communication, respectively. Third, a fixed-time distributed average tracking observer for double-integrator-type multiagent systems is designed to measure the average value of multiple reference signals under undirected communication. Note that all the proposed observers are constructed with time-based generators and can be trivially extended to that for high-order integrator-type multiagent systems. Furthermore, by combining the proposed fixed-time sliding-mode control method with the information provided by the fixed-time observers, the fixed-time controllers are designed to solve the fixed-time distributed consensus tracking and the distributed average tracking problems. Finally, a few numerical simulations are shown to verify the results.

Index Terms—Consensus tracking, distributed average tracking, distributed observer, fixed time, sliding-mode control, time-based generator.

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Qiang Chen is with the School of Computer Science, the School of Artificial Intelligence, Optics and Electronics (iOPEN), the School of Electronics and Information, and the School of Automation, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: chenq@mail.nwpu.edu.cn).

Yu Zhao is with the School of Automation, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: yuzhao5977@gmail.com).

Guanghui Wen is with the School of Mathematics, Southeast University, Nanjing 210096, China (e-mail: wenguanghui@gmail.com).

Guoqing Shi is with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: shiguogqing@nwpu.edu.cn).

Xinghuo Yu is with the School of Engineering, RMIT University, Melbourne, VIC 3001, Australia (e-mail: x.yu@rmit.edu.au).

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I. INTRODUCTION

DISTRIBUTED cooperation control has been a popular research issue over the past decades due to its significant value in reality, such as distributed optimization [1], [2]; tracking control [3], [4], [5]; and flocking and containment control [6], [7], [8].

In distributed cooperation control of a flock of agents, an important research issue is to develop an algorithm that makes the agents achieve consensus. The consensus algorithm for single-integrator multiagent systems was first developed in [9], and then some sufficient and necessary conditions for the consensus of double-integrator multiagent systems were generalized in [10]. Distributed tracking control can be regarded as an extension of generalized consensus control, in which the followers not only have to reach consensus but also to follow with the specified trajectory. For example, in distributed consensus tracking and distributed average tracking, the target trajectories are the states of the leader and the average value of multiple reference signals, respectively. However, in distributed control systems, only a few of the agents can directly acquire the target information. Then, to obtain the target information, resorting to observers is a frequently used method. Some observer-based algorithms for solving the distributed consensus tracking problem were proposed in [11], [12], and [13]. For solving the distributed average tracking problem, some observers were developed in [14], [15], [16], and [17].

However, these above observers are asymptotically stable, which implies that the upper bounded convergence time is not guaranteed. Compared with the asymptotic control approach, the finite-time control technique usually has a faster response. Therefore, many finite-time control methods have been developed during the past few years [18], [19], [20], [21]. Moreover, some novel finite-time observers for multiagent systems were proposed in [22], [23], [24], and [25]. Nevertheless, the upper bounded convergence time of finite-time algorithms depends on the initial states of agents. In some engineering practices, the initial states are not available and hence the upper bounded convergence time can not be guaranteed as well. Motivated by this fact, the fixed-time stability strategy was first investigated in [26], in which the predefined convergence time does not depend on system initial conditions. Then, some novel fixed-time algorithms for single-integrator

multiagent systems were developed in [27]. For the double-integrator-type multiagent system under the undirected communication topology, a kind of fixed-time observer was designed to estimate the states of the leader in [28]. Then, Tian et al. developed a fixed-time method without relying velocity measurement to solve the similar problem in [29]. Under the directed communication, a kind of fixed-time attractive observer for high-order integrator-type multiagent systems was developed in [30]. Note that this algorithm can only guarantee the system states to converge to a specific attractive region in the predefined time, and then the algorithm becomes asymptotically stable. To reduce energy consumption, Ni et al. [31] proposed to introduce the event-triggered strategy into the fixed-time consensus tracking algorithm for high-order multiagent systems.

After the observer obtains the object trajectory in a fixed-time, the fixed-time controller will be used to drive the agent to track the object trajectory. For the double-integrator-type control system, sliding-mode control method is a classic nonlinear control protocol [32], which has the advantages of fast response, parameter change, insensitive to disturbance and simple physical implementation. Although some finite-time sliding-mode control methods have been proposed in [33] and [34], extending them to the fixed-time algorithm for double-integrator-type control systems is nontrivial. Zuo proposed a fixed-time sliding-mode control protocol by utilizing the sinusoid function to offset the singularity in the neighborhood of zero [35], but it lacks consideration to the disturbances. Moreover, a similar fixed-time convergent sliding surface algorithm can be found in [36].

The common ground of the above fixed-time protocol is that they incorporate power functions into their control inputs, which are prone to result in system input saturation. Recently, Ning et al. [37] proposed a novel fixed-time consensus strategy for single-integrator multiagent systems by using time-based generators. Compared with the power function based methods, the time-based generator approaches need a less magnitude of control input and it could hence effectively avoid system input saturation. Some related references about time-based generators can be found in [38], [39], and [40]. For the time synchronization between different agents, the clock synchronization device has been proposed to guarantee the time synchronization [41], [42], and it is hence not reiterated here.

Compared with the above related works, five main contributions are made in this article. First, a novel fixed-time sliding-mode control method is developed. Compared with the fixed-time algorithm in [35] and [36], its control input does not include the power function that might result in system input saturation. Moreover, the disturbance is taken into consideration. Second, a novel fixed-time distributed observer for second-order tracking problems under the undirected topology is proposed. Compared with the fixed-time observer in [28] and [29], the predefined convergence time of the observer via time-based generators can be adjusted more easily. Third, the fully fixed-time distributed consensus tracking observer under directed communication is proposed. Compared with the prior work in [30], the proposed method can precisely design the upper bounded convergence time independent of

initial system states and the system disturbance is analyzed. Fourth, the fixed-time observer for distributed average tracking problems under the undirected topology graph is designed. Compared with the finite-time observer in [25], the proposed method can achieve fixed-time stability. Moreover, to the best of our knowledge, the second-order fixed-time observer for distributed average tracking under the undirected topology graph is first proposed in this article. Furthermore, all the proposed observers can be easily extended to that for high-order integrator-type control systems. Finally, by combining the proposed sliding-mode control protocol with the proposed observers, three fixed-time controllers are developed to solve the fixed-time distributed consensus tracking problem under the undirected graph and directed graph, and the fixed-time distributed average tracking problem under the undirected graph, respectively.

Furthermore, the proposed algorithms in this article have a broad range of potential applications. The sliding-mode control algorithm has been used in the control of vehicles, spacecrafts, and robot manipulators [39], [43]. The distributed consensus tracking algorithms have been widely applied to aerospace systems [44], mechanical systems [45], and dynamic region following [46].

The remainder of this article is given as follows. In Section II, some mathematical preliminaries are given. In Section III, the fixed-time sliding-mode control protocol is proposed. Next, the fixed-time observers for distributed consensus tracking and distributed average tracking problems are designed. Then, the controllers for fixed-time distributed consensus tracking and fixed-time distributed average tracking problems are given. In Section IV, several simulations are given. In Section V, a few conclusions are made.

II. MATHEMATICAL PRELIMINARIES

A. Notations

The real number set and the N -dimensional real vector space are denoted by $\mathbb{R}$ and $\mathbb{R}^n$, respectively. The signum function is represented by $\text{sgn}(\cdot)$ and its vector form can be written as $\text{sgn}(z) = [\text{sgn}(z_1), \text{sgn}(z_2), \dots, \text{sgn}(z_n)]^T$, where $z = [z_1, z_2, \dots, z_n]^T$. Let $|\cdot|$ stand for the absolute value of a scalar. The q -norm of vector z can be written as $\|z\|_q = (|z_1|^q + |z_2|^q + \dots + |z_n|^q)^{1/q}$. Let $\lambda_1(Q)$ and $\lambda_2(Q)$ represent the smallest and the second-smallest eigenvalue of matrix Q , respectively.

B. Graph Theory

The communication topology of a group of $n + 1$ agents can be represented by a graph $\mathcal{G}$. If there is a leader in them, the other n agents can be expressed as a subgraph $\mathcal{G}_s$. The weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed with a set of nodes $\mathcal{V} = \{v_1, v_2, \dots, v_{n+1}\}$ and a set of edges $\mathcal{E} = \{e_1, e_2, \dots, e_m\}$. A directed edge from v_j to v_i can be denoted as (v_i, v_j) , which means v_i can receive information from v_j . A directed path from v_j to v_i consists of a sequence of edges in the form of $\mathcal{E}_{ij} = \{(v_i, v_{i+1}), \dots, (v_{j-1}, v_j)\}$, which means the information can flow from v_j to v_i . When replacing the directed edges by the undirected, it becomes undirected

path and the information flow is bidirectional. It is said to contain a spanning tree if at least there exists a node which has directed paths to all other nodes. The undirected graph is connected if and only if there at least exists an undirected path between any two nodes. Let $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ and $D \in \mathbb{R}^{n \times m}$ denote the adjacency matrix and the incidence matrix of the graph, respectively, and $a_{ij} = 1$ if there exists a directed edge from v_j to v_i , else $a_{ij} = 0$. With regard to undirected graphs, $a_{ij} = a_{ji}$. let $O = [o_{ij}] \in \mathbb{R}^{n \times n}$ denote the degree matrix, where $o_{ii} = \sum_{j=1}^n a_{ij}$ and other $o_{ij} = 0$. Then, the Laplacian matrix is written as $L \in \mathbb{R}^{n \times n} = O - A$. Set $a_{i0} = 1$ if the agent i can acquire information from the leader, else $a_{i0} = 0$, and then set $B = \text{diag}\{a_{i0}, a_{i1}, \dots, a_{in}\}$.

C. Time-Based Generator

The time-based generator $\xi(t)$ is a kind of time-dependent function that can be seen as a termination function. Its general properties can be generalized as follows.

- 1) $\xi(t)$ is a nondecreasing and continuous function.
- 2) With time growing, $\xi(t)$ increases from the initial state $\xi(0) = 0$ to $\xi(t_s) = 1$, and when $t > t_s$, $\xi(t) \equiv 1$, where t_s can be predesigned arbitrarily.
- 3) $\dot{\xi}(0) = 0$ and when $t \geq t_s$, $\dot{\xi}(t) \equiv 0$.

Remark 1: A typical time-based generator function $\xi(t)$ is presented as follows [37]:

$$\begin{cases} \xi(t) = \frac{10}{t_s^6} t^6 - \frac{24}{t_s^5} t^5 + \frac{15}{t_s^4} t^4, & 0 \leq t \leq t_s \\ 1, & t > t_s. \end{cases}$$

Consider the differential equation as follows:

$$\dot{z} = -h(t)z, \quad z(0) = z_0 \quad (1)$$

where $h(t)$ is constructed as

$$h(t) = k \frac{\dot{\xi}(t)}{1 - \xi(t) + \delta} \quad (2)$$

where $k \in \mathbb{R}$ and $\delta \in \mathbb{R}$ are two positive constants that satisfy $k > 1$ and $0 < \delta < 1$.

By solving (1), one has

$$z = \left(\frac{1 - \xi(t) + \delta}{1 + \delta} \right)^k z_0. \quad (3)$$

With time t grows from 0 to t_s , the time-based generator function $\xi(t)$ increases from 0 to 1 smoothly. Therefore, when $t \in [0, t_s]$, the system variable z gradually approaches to $z_0(\delta/1 + \delta)^k$. When $t \geq t_s$, the system variable z stays at the fixed point $z_0(\delta/1 + \delta)^k$. If let $\delta = 0.001$ and $k = 3$, when $t \geq t_s$, the solution of (1) will be $z \approx 10^{-9} z_0$. For this reason, we can nearly think that z reaches to zero at t_s and the initial state z_0 has no effect on the convergence time.

D. Problem Description

1) *Fixed-Time Sliding-Mode Control:* A typical double-integrator-type control system is given as follows:

$$\begin{cases} \dot{z}_1 = z_2 \\ \dot{z}_2 = u + \varrho \end{cases} \quad (4)$$

where $z_1 \in \mathbb{R}$ and $z_2 \in \mathbb{R}$ denote the position and velocity, respectively. Variable $\varrho \in \mathbb{R}$ is a disturbance bounded by a positive constant $\varrho_{\max}$.

The objective of fixed-time sliding-mode control is to devise a control input u that drives system (4) to the equilibrium point in a fixed time. In this article, the definition of fixed-time stability for system states (4) is given as follows.

Definition 1: For system (4), it is said to achieve fixed-time stability if the system satisfies

$$\begin{cases} \lim_{t \rightarrow T_{\max}} |z_1| + |z_2| \leq c \\ \lim_{t \rightarrow \infty} |z_1| + |z_2| = 0 \end{cases} \quad (5)$$

where $T_{\max}$ is the upper bounded convergence time independent of initial system states, and c is a small positive constant close to zero.

2) *Fixed-Time Distributed Consensus Tracking:* Suppose that there is a double-integrator-type multiagent system with a leader and n followers. The leader can be modeled by

$$\begin{cases} \dot{x}_0 = v_0 \\ \dot{v}_0 = u_0 \end{cases} \quad (6)$$

where $x_0 \in \mathbb{R}$ and $v_0 \in \mathbb{R}$ represent the position and velocity of the leader, respectively, and $u_0 \in \mathbb{R}$ represents the control input that is bounded by a positive constant $u_{\max}$.

Then, the followers can be modeled by:

$$\begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = u_i + d_i, \quad i = 1, 2, \dots, n \end{cases} \quad (7)$$

where $x_i \in \mathbb{R}$ and $v_i \in \mathbb{R}$ denote the position and velocity of agent i , respectively. $u_i \in \mathbb{R}$ denotes the control input and $d_i \in \mathbb{R}$ denotes the disturbance satisfying $|d_i| \leq d_{\max}$.

The objective of fixed-time distributed consensus tracking is to devise the control input for each follower that only depends on the local information. Meanwhile, the control inputs enable the followers to achieve consensus with the leader in a fixed time independent of initial system states.

Definition 2 (Fixed-Time Distributed Consensus Tracking): For the system described by (6) and (7), with the given observer and control input u_i , it is said to achieve fixed-time distributed consensus tracking if all the followers can achieve consensus with the leader in a fixed-time $T_{\max}$ independent of initial system states, that is

$$\begin{cases} \lim_{t \rightarrow T_{\max}} |x_i - x_0| + |v_i - v_0| \leq c \\ \lim_{t \rightarrow \infty} |x_i - x_0| + |v_i - v_0| = 0 \end{cases} \quad (8)$$

where $T_{\max}$ can be predesigned arbitrarily independent of initial conditions and c can be limited to the desired level.

3) *Fixed-Time Distributed Average Tracking:* Consider a double-integrator-type multiagent system with n agents modeled by (7). Each agent i has a reference signal $r_i \in \mathbb{R}$ described as follows:

$$\begin{cases} \dot{r}_i = f_i \\ \dot{f}_i = a_i^r, \quad i = 1, 2, \dots, n \end{cases} \quad (9)$$

where f_i and a_i^r are the velocity and acceleration of reference signal r_i , respectively. Note that a_i^r is bounded by a positive constant $a_{\max}$. Let $\bar{r} = (1/n) \sum_{i=1}^n r_i$, $\bar{f} = (1/n) \sum_{i=1}^n f_i$, and $\bar{a}^r = (1/n) \sum_{i=1}^n a_i^r$ denote the average values of the reference signals.

The objective of fixed-time distributed average tracking is to devise the control input for each agent that only depends on the local information. Meanwhile, the control inputs enables the agents to achieve consensus with the average value of the multiple reference signals in a fixed time without dependence on initial system states.

Definition 3 (Fixed-Time Distributed Average Tracking): For the system described by (7) and (9), with the given observer and control input u_i , it is said to achieve fixed-time distributed average tracking if all the agents can achieve consensus with the average value of the multiple reference signals in a fixed-time $T_{\max}$ independent of initial states, that is

$$\begin{cases} \lim_{t \rightarrow T_{\max}} |x_i - \bar{r}| + |v_i - \bar{f}| \leq c \\ \lim_{t \rightarrow \infty} |x_i - \bar{r}| + |v_i - \bar{f}| = 0. \end{cases} \quad (10)$$

III. MAIN RESULTS

A. Fixed-Time Sliding-Mode Control

Lemma 1 [30]: Suppose that $z(0) = z_0$ and $V(z)$ is a positive-definite Lyapunov candidate that satisfies the inequality as follows:

$$\dot{V}(z) + \mu V^\nu(z) \leq 0 \quad (11)$$

where $\mu \geq 0$ and $\nu \in (0, 1)$. Then, z will converge to zero in a finite time $T(z_0)$ such that

$$T(z_0) \leq \frac{1}{\mu(1-\nu)} V^{1-\nu}(z_0). \quad (12)$$

In this article, Lemma 1 is mainly used to demonstrate that the system states still stay at the equilibrium after the predefined convergence time.

The process of fixed-time double-integrator sliding-mode control is generally divided into two sections. In the first section, the control input forces the system states to arrive on the prescribed surface in a fixed time t_{s1} ; in the second section, the system states will slide along the surface to the equilibrium point in a fixed time t_{s2} . Therefore, the entire convergence time is bounded by $T_{\max} = t_{s2}$. In order to force the system states to arrive at the specific positions in the fixed time during each stage, two time-based generators $\xi_1(t)$ and $\xi_2(t)$ are used sequentially. The detail of $\xi_1(t)$ and $\xi_2(t)$ is shown as follows:

$$\xi(t) = \begin{cases} \xi_1(t), & 0 \leq t < t_{s1} \\ \xi_2(t), & t_{s1} \leq t < t_{s2} \\ 1, & t_{s2} \leq t < \infty. \end{cases} \quad (13)$$

According to (2), the corresponding $h(t)$ is constructed as follows:

$$h(t) = \begin{cases} h_1(t) = k \frac{\dot{\xi}_1(t)}{1 - \xi_1(t) + \delta_1}, & 0 \leq t < t_{s1} \\ h_2(t) = k \frac{\dot{\xi}_2(t)}{1 - \xi_2(t) + \delta_2}, & t_{s1} \leq t < t_{s2} \\ 0, & t_{s2} \leq t < \infty. \end{cases} \quad (14)$$

Remark 2: Since $\dot{\xi}_1(0) = \dot{\xi}_1(t_{s1}) = \dot{\xi}_2(t_{s1}) = \dot{\xi}_2(t_{s2}) = 0$, one obtains $h_1(0) = h_1(t_{s1}) = h_2(t_{s1}) = h_2(t_{s2}) = 0$, which shows the continuity of $h(t)$. Moreover, due to the nonnegativity of $\dot{\xi}_1(t)$ and $\dot{\xi}_2(t)$, function $h(t)$ is non-negative.

Remark 3: In order to clarify the idea clearly, we directly let $\xi_1(t)$ and $\xi_2(t)$ have the same structure in the proof. But in simulation, due to the discontinuity of $\xi(t)$, that is,

$\xi_1(t_{s1}) = 1$ and $\xi_2(t_{s1}) = 0$, the sharp decrease may lead to the unavailability of the derivative of $\xi_2(t_{s1})$. One method to solve this problem is to reset $\tilde{\xi}_2(t) = \xi_2(t) + 1$ and $\tilde{h}_2(t) = k([\tilde{\xi}_2(t)]/[2 - \tilde{\xi}_2(t) + \delta_2])$, respectively.

For system (4), the fixed-time sliding-mode surface is designed as follows:

$$s = \left(\frac{1}{2}h(t) + 1\right)z_1 + z_2. \quad (15)$$

If $s = 0$, the system states arrive at the sliding-mode surface and then can be rewritten as follows:

$$z_2 = \dot{z}_1 = -\left(\frac{1}{2}h(t) + 1\right)z_1. \quad (16)$$

The control input for system (4) is devised as follows:

$$\begin{aligned} u = & -\frac{1}{2}\dot{h}(t)z_1 - \left(\frac{1}{2}h(t) + 1\right)z_2 \\ & - \frac{1}{2}h(t)s - \rho \operatorname{sgn}(s) \end{aligned} \quad (17)$$

where ρ is a positive constant satisfying $\rho \geq \varrho_{\max} + 1$.

Theorem 1: With the given control input (17), system (4) will achieve the fixed-time stability (see Definition 1) within the upper bounded convergence time $T_{\max} = t_{s2}$.

Proof: The Lyapunov candidate is constructed as $V_1 = (1/2)s^2$. Differentiate (15) against time, and then one has

$$\dot{s} = \frac{1}{2}\dot{h}(t)z_1 + \left(\frac{1}{2}h(t) + 1\right)z_2 + u + \varrho. \quad (18)$$

Substitute control input (17) into (18), and then one has

$$\dot{s} = -\frac{1}{2}h(t)s - \rho \operatorname{sgn}(s) + \varrho. \quad (19)$$

Differentiate the Lyapunov candidate V_1 against time, and substitute (19) into V_1 , and then one has

$$\begin{aligned} \dot{V}_1 &= s\dot{s} \\ &= -\frac{1}{2}h(t)s^2 - \rho|s| + \varrho s \\ &\leq -\frac{1}{2}h(t)s^2 - (\rho - \varrho_{\max})|s| \\ &\leq -\frac{1}{2}h(t)s^2 \\ &= -h(t)V_1 \end{aligned} \quad (20)$$

where the second inequality holds due to the precondition $\rho \geq \varrho_{\max} + 1$.

When $t \in [0, t_{s1})$, $h(t) = h_1(t)$. According to the property of (1) one obtains

$$\begin{aligned} \lim_{t \rightarrow t_{s1}} V_1 &\leq \left(\frac{1 - \xi_1(t_{s1}) + \delta_1}{1 + \delta_1}\right)^k V_1(0) \\ &= \left(\frac{\delta_1}{1 + \delta_1}\right)^k V_1(0) \end{aligned} \quad (21)$$

where the value of $(\delta_1/1 + \delta_1)^k V_1(0)$ is very small and close to zero, which also means that the system states are in the near region of the sliding-mode surface.

When $t \geq t_{s1}$, $h(t) = h_2(t) \geq 0$ and then one obtains

$$\begin{aligned}\dot{V}_1 &= -\frac{1}{2}h_2(t)s^2 - \rho|s| + \varrho s \\ &\leq -(\rho - \varrho_{\max})|s| \\ &\leq -|s| \\ &= -\sqrt{2V_1}.\end{aligned}\quad (22)$$

According to Lemma 1, one obtains that V_1 will converge to zero after t_{s1} within a finite time $\hat{t}_{s1}$, where $\hat{t}_{s1} \leq \sqrt{2V_1(t_{s1})} \leq \sqrt{2V_1(0)(\delta_1/1 + \delta_1)^k}$. Although V_1 may not converge to zero perfectly at t_{s1} , which means that the system states are in the adjacent region of the sliding-mode surface ($s = 0$), we will prove the system states will still converge to zero along the sliding surface in the fixed-time t_{s2} . A Lyapunov candidate is constructed as $V_2 = (1/2)z_1^2$. The proof will be given in both cases: 1) V_1 perfectly converges to zero at t_{s1} and 2) V_1 converges to the adjacent region of zero at t_{s1} .

Case 1: $V_1(t_{s1}) = 0$.

Since $V_1(t_{s1}) = 0$, one has $s = 0$. Differentiate V_2 along (16), and then one obtains

$$\begin{aligned}\dot{V}_2 &= z_1 \dot{z}_1 \\ &= -\frac{1}{2}h(t)z_1^2 - z_1^2 \\ &\leq -\frac{1}{2}h(t)z_1^2 \\ &= -h(t)V_2.\end{aligned}\quad (23)$$

When $t \in [t_{s1}, t_{s2})$, one has $h(t) = h_2(t)$. Therefore, according to (1), one obtains $\lim_{t \rightarrow t_{s2}} V_2 \leq (\delta_2/1 + \delta_2)^k V_2(t_{s1})$, which means the system states are in the adjacent region of zero, that is, $\lim_{t \rightarrow t_{s2}} |z_1| \leq \sqrt{2(\delta_2/1 + \delta_2)^k V_2(t_{s1})}$. When $t \geq t_{s2}$, due to $h(t) = 0$ and $\dot{V}_2 = -z_1^2 = -2V_2$, one can conclude that V_2 and z_1 will converge to zero exponentially. Moreover, since when $t \geq t_{s2}$, $z_2 = -(h(t) + 1)z_1 = -z_1$, system state z_2 will also converge to zero with the same rate of z_1 .

Case 2: $V_1(t_{s1}) \neq 0$.

According to the relationship between V_1 and s , suppose that the system states converge to the adjacent region of the sliding surface at t_{s1} and there exists an error e , that is, $s = \{e|t \geq t_{s1}\}$. When $t \geq t_{s1}$, according to (22), one has $\dot{V}_1 \leq -\sqrt{2V_1}$, which means V_1 as well as $|e|$ are nonincreasing. Meanwhile, they are bounded by $\hat{V}_1 = (\delta_1/1 + \delta_1)^k V_1(0)$ and $|\hat{e}| = \sqrt{2(\delta_1/1 + \delta_1)^k V_1(0)}$, respectively. Then, sliding-mode surface (15) can be rewritten as

$$e = \left(\frac{1}{2}h(t) + 1\right)z_1 + z_2. \quad (24)$$

The derivative of z_1 against time can be written as follows:

$$\dot{z}_1 = -\left(\frac{1}{2}h(t) + 1\right)z_1 + e. \quad (25)$$

Substitute (25) into $\dot{V}_2$, and then one has

$$\begin{aligned}\dot{V}_2 &= -\frac{1}{2}h(t)z_1^2 - z_1^2 + ez_1 \\ &= -h(t)V_2 - z_1^2 + ez_1.\end{aligned}\quad (26)$$

Note that $|e|$ is very small and converges to zero quickly, and the upper bounded convergence time is bounded by $\hat{t}_{s1}$.

If $|z_1| < |\hat{e}|$ at t_{s1} , although we may have $-z_1^2 + ez_1 \geq 0$, the value of $|z_1|$ has been in the adjacent region of zero. Meanwhile, we have $|z_1| < |\hat{e}| \leq \sqrt{2(\delta_1/1 + \delta_1)^k V_1(0)}$, because $z_1 \geq |e|$ leads to $\dot{V}_2 \leq 0$, which means z_1 will move to the direction of zero. After $t_{s1} + \hat{t}_{s1}$, due to $|e| = 0$, $|z_1|$ will at least converge exponentially.

If $|z_1| \geq |e|$ at t_{s1} , one has $\dot{V}_2 \leq -h(t)V_2$. According to (1), one can obtain $\lim_{t \rightarrow t_{s2}} V_2 \leq (\delta_2/1 + \delta_2)^k V_2(t_{s1})$, which means V_2 converges to the adjacent region of zero at t_{s2} , that is, $\lim_{t \rightarrow t_{s2}} |z_1| \leq \sqrt{2(\delta_2/1 + \delta_2)^k V_2(t_{s1})}$. Therefore, one can conclude that the margin of $|z_1|$ at t_{s2} is bounded by $\sqrt{2(\delta_1/1 + \delta_1)^k V_1(0)}$ or $\sqrt{2(\delta_2/1 + \delta_2)^k V_2(t_{s1})}$. When $t \geq t_{s2}$, due to $z_1 = -z_2$, both z_1 and z_2 will converge to zero exponentially.

Therefore, in both cases, V_2 will converge nearly to zero in a fixed time $T_{\max} = t_{s2}$ and then converge quickly to zero. Therefore, the proof has been completed. ■

B. Fixed-Time Distributed Consensus Tracking Observer Under Undirected Communication

Assumption 1: The topology subgraph $\mathcal{G}_s$ for the followers is undirected and connected; there at least exists a follower which can acquire information from the leader.

Lemma 2 [28]: If $L \in \mathbb{R}^{n \times n}$ is the Laplacian matrix of a undirected connected graph, and the non-negative diagonal matrix $B = \text{diag}\{a_{10}, \dots, a_{n0}\} \in \mathbb{R}^{n \times n}$ with at least one element greater than zero, then $Q = L + B$ is a positive-definite matrix.

In the section, a fixed-time distributed observer under dynamics (6) and (7) is designed for each follower to measure the relative position and velocity disagreements between the leader and itself under undirected communication. Denote the real tracking errors as $\tilde{x}_i = x_i - x_0$ and $\tilde{v}_i = v_i - v_0$. Then, fixed-time distributed observers α_i and β_i are developed to estimate $\tilde{x}_i$ and $\tilde{v}_i$. The specific formulation is shown as follows:

$$\begin{aligned}\dot{\alpha}_i &= \beta_i - \eta(t) \left\{ \sum_{j=0}^n a_{ij} [(\alpha_i - \alpha_j) - (x_i - x_j)] \right\} \\ &\quad - b_1 \text{sgn} \left\{ \sum_{j=0}^n a_{ij} [(\alpha_i - \alpha_j) - (x_i - x_j)] \right\} \\ \dot{\beta}_i &= u_i - \eta(t) \left\{ \sum_{j=0}^n a_{ij} [(\beta_i - \beta_j) - (v_i - v_j)] \right\} \\ &\quad - c_1 \text{sgn} \left\{ \sum_{j=0}^n a_{ij} [(\beta_i - \beta_j) - (v_i - v_j)] \right\}\end{aligned}\quad (27)$$

where $i = 1, \dots, n$, $\eta(t) = (1/[2\lambda_1(Q)])h(t)$, $\alpha_0 = 0$, and $\beta_0 = 0$. b_1 and c_1 are positive constants satisfying $b_1 \geq 1$ and $c_1 > u_{\max} + d_{\max}$.

Theorem 2: With the given dynamics (6) and (7) and observer (27), under Assumption 1, α_i and β_i converge to $\tilde{x}_i$ and $\tilde{v}_i$ within a fixed-time $T_{\max} = t_{s2}$, that is

$$\begin{cases} \lim_{t \rightarrow t_{s1}} |\beta_i - \tilde{v}_i| \leq o_1 \\ \lim_{t \rightarrow t_{s2}} |\alpha_i - \tilde{x}_i| \leq o_2 \\ \lim_{t \rightarrow \infty} |\beta_i - \tilde{v}_i| = 0 \\ \lim_{t \rightarrow \infty} |\alpha_i - \tilde{x}_i| = 0 \end{cases}\quad (28)$$

where t_{s1} and t_{s2} are the upper bounded convergence time [see (14)]. Parameters o_1 and o_2 are small positive constants close to zero and can be limited to a desired level.

Proof: Let $\tilde{\alpha}_i = \alpha_i - \tilde{x}_i$ and $\tilde{\beta}_i = \beta_i - \tilde{v}_i$ be the errors between the observed disagreements and the real disagreements. According to (27), $\dot{\tilde{\alpha}}_i$ and $\dot{\tilde{\beta}}_i$ can be written as

$$\begin{aligned}\dot{\tilde{\alpha}}_i &= \tilde{\beta}_i - \eta(t) \sum_{j=0}^n a_{ij}(\tilde{\alpha}_i - \tilde{\alpha}_j) \\ &\quad - b_1 \operatorname{sgn} \left[\sum_{j=0}^n a_{ij}(\tilde{\alpha}_i - \tilde{\alpha}_j) \right] \\ \dot{\tilde{\beta}}_i &= -\eta(t) \sum_{j=0}^n a_{ij}(\tilde{\beta}_i - \tilde{\beta}_j) \\ &\quad - c_1 \operatorname{sgn} \left[\sum_{j=0}^n a_{ij}(\tilde{\beta}_i - \tilde{\beta}_j) \right] - d_i + u_0.\end{aligned}\quad (29)$$

Let $\tilde{\alpha} = [\tilde{\alpha}_1, \dots, \tilde{\alpha}_n]^T$, $\tilde{\beta} = [\tilde{\beta}_1, \dots, \tilde{\beta}_n]^T$, $d = [d_1, \dots, d_n]^T$ and $u \in \mathbb{R}^n = [u_0, \dots, u_0]^T$. The vector form of (29) can be written as

$$\begin{aligned}\dot{\tilde{\alpha}} &= \tilde{\beta} - \eta(t) Q \tilde{\alpha} - b_1 \operatorname{sgn}(Q \tilde{\alpha}) \\ \dot{\tilde{\beta}} &= -\eta(t) Q \tilde{\beta} - c_1 \operatorname{sgn}(Q \tilde{\beta}) - d + u\end{aligned}\quad (30)$$

where $Q = L + B$ is a positive-definite matrix according to Lemma 2, matrix L is the Laplacian matrix for the followers, and matrix $B = \operatorname{diag}\{a_{10}, \dots, a_{n0}\}$ represents the communication topology between the leader and the followers. For example, if the i th agent can receive information from the leader, $a_{i0} = 1$; else, $a_{i0} = 0$.

Construct a Lyapunov candidate as $V_3 = (1/2) \tilde{\beta}^T Q \tilde{\beta}$. Since Q is a positive-definite matrix, V_3 is well defined. Differentiate V_3 against time such that

$$\begin{aligned}\dot{V}_3 &= \tilde{\beta}^T Q \dot{\tilde{\beta}} \\ &= \tilde{\beta}^T Q \left[-\eta(t) Q \tilde{\beta} - c_1 \operatorname{sgn}(Q \tilde{\beta}) - d + u \right] \\ &\leq -\eta(t) \left(Q^{\frac{1}{2}} \tilde{\beta} \right)^T Q \left(Q^{\frac{1}{2}} \tilde{\beta} \right) \\ &\quad - (c_1 - u_{\max} - d_{\max}) \|Q \tilde{\beta}\|_1.\end{aligned}\quad (31)$$

Due to the precondition $c_1 > u_{\max} + d_{\max}$, one can obtain

$$\begin{aligned}\dot{V}_3 &\leq -\eta(t) \left(Q^{\frac{1}{2}} \tilde{\beta} \right)^T Q \left(Q^{\frac{1}{2}} \tilde{\beta} \right) \\ &\leq -\lambda_1(Q) \eta(t) \tilde{\beta}^T Q \tilde{\beta} \\ &= -h(t) V_3.\end{aligned}\quad (32)$$

When $t \in [0, t_{s1})$, $h(t) = h_2(t)$. According to (1) one can conclude that

$$\begin{aligned}\lim_{t \rightarrow t_{s1}} V_3 &= \lim_{t \rightarrow t_{s1}} \frac{1}{2} \tilde{\beta}^T (L + B) \tilde{\beta} \\ &= \lim_{t \rightarrow t_{s1}} \frac{1}{4} \sum_{i=1}^n \sum_{j=1}^n a_{ij} (\tilde{\beta}_i - \tilde{\beta}_j)^2 + \frac{1}{2} \sum_{i=1}^n a_{i0} \tilde{\beta}_i^2 \\ &\leq \left(\frac{\delta_1}{1 + \delta_1} \right)^k V_3(0).\end{aligned}\quad (33)$$

According to inequality $(1/2) \sum_{i=1}^n a_{i0} \tilde{\beta}_i^2 \leq (\delta_1/1 + \delta_1)^k V_3(0)$, one can obtain $\lim_{t \rightarrow t_{s1}} |\tilde{\beta}_i| \leq$

$\sqrt{2(\delta_1/1 + \delta_1)^k V_3(0)}$. Therefore, the velocity observer β_i successfully observe the relative velocity $v_i - v_0$ in the fixed time t_{s1} .

When $t \geq t_{s1}$, according to (31), one has

$$\dot{V}_3 \leq -(c_1 - u_{\max} - d_{\max}) \|Q \tilde{\beta}\|_1. \quad (34)$$

Due to

$$\|Q \tilde{\beta}\|_1 \geq \|Q \tilde{\beta}\|_2 = \sqrt{(Q \tilde{\beta})^T Q \tilde{\beta}} \geq \sqrt{\lambda_1(Q) \tilde{\beta}^T Q \tilde{\beta}} \quad (35)$$

one obtains

$$\dot{V}_3 \leq -(c_1 - u_{\max} - d_{\max}) \sqrt{2\lambda_1(Q) V_3} \leq 0. \quad (36)$$

Following from Lemma 1, when $t \geq t_{s1}$, V_3 will quickly converge to zero within an upper bounded convergence time $\hat{t}_{s1} \leq ([\sqrt{2V_3(t_{s1})/\lambda_1(Q)}]/[c_1 - u_{\max} - d_{\max}]) \leq ([\sqrt{2(\delta_1/1 + \delta_1)^k V_3(0)}/\lambda_1(Q)]/[c_1 - u_{\max} - d_{\max}])$. Although V_3 may not converge perfectly to zero at t_{s1} , the fixed-time stability of position observer α_i is not affected. The proof of position observer α_i is given as follows.

Construct a Lyapunov candidate as $V_4 = (1/2) \tilde{\alpha}^T Q \tilde{\alpha}$. Differentiate it against time and then one has

$$\begin{aligned}\dot{V}_4 &= \tilde{\alpha}^T Q \dot{\tilde{\alpha}} \\ &= \tilde{\alpha}^T Q \tilde{\beta} - \eta(t) \tilde{\alpha}^T Q Q \tilde{\alpha} - b_1 \tilde{\alpha}^T Q \operatorname{sgn}(Q \tilde{\alpha}) \\ &= -\eta(t) \tilde{\alpha}^T Q Q \tilde{\alpha} - b_1 \|Q \tilde{\alpha}\|_1 + (Q \tilde{\alpha})^T \tilde{\beta}.\end{aligned}\quad (37)$$

Since when $t \geq t_{s1}$, one has $|\tilde{\beta}_i| \leq \sqrt{2(\delta_1/1 + \delta_1)^k V_3(0)} < 1$. Then, due to the precondition $b_1 \geq 1$, one can conclude

$$\begin{aligned}\dot{V}_4 &\leq -\lambda_1(Q) \eta(t) \tilde{\alpha}^T Q \tilde{\alpha} - (b_1 - 1) \|Q \tilde{\alpha}\|_1 \\ &\leq -\lambda_1(Q) \eta(t) \tilde{\alpha}^T Q \tilde{\alpha} \\ &= -h(t) V_4.\end{aligned}\quad (38)$$

When $t \in [t_{s1}, t_{s2})$, $h(t) = h_2(t)$. According to (1), one concludes that

$$\begin{aligned}\lim_{t \rightarrow t_{s2}} V_4 &= \lim_{t \rightarrow t_{s2}} \frac{1}{2} \tilde{\alpha}^T (L + B) \tilde{\alpha} \\ &= \lim_{t \rightarrow t_{s2}} \frac{1}{4} \sum_{i=1}^n \sum_{j=1}^n a_{ij} (\tilde{\alpha}_i - \tilde{\alpha}_j)^2 + \frac{1}{2} \sum_{i=1}^n a_{i0} \tilde{\alpha}_i^2 \\ &\leq \left(\frac{\delta_2}{1 + \delta_2} \right)^k V_4(t_{s1}).\end{aligned}\quad (39)$$

According to the inequality $(1/2) \sum_{i=1}^n a_{i0} \tilde{\alpha}_i^2 \leq (\delta_2/1 + \delta_2)^k V_4(t_{s1})$, one can obtain $\lim_{t \rightarrow t_{s2}} |\tilde{\alpha}_i| \leq \sqrt{2(\delta_2/1 + \delta_2)^k V_4(t_{s1})}$. Therefore, position observer α_i successfully observe the relative position $x_i - x_0$ in the fixed time $T_{\max} = t_{s2}$.

When $t \geq t_{s2}$, according to (35) and (38), one has

$$\begin{aligned}\dot{V}_4 &\leq -(b_1 - 1) \|Q \tilde{\beta}\|_1 \\ &\leq -(b_1 - 1) \sqrt{2\lambda_1(Q) V_4}.\end{aligned}\quad (40)$$

Following from Lemma 1, when $t \geq t_{s2}$, V_4 will quickly converge to zero within an upper bounded convergence time $\hat{t}_{s2} \leq \frac{\sqrt{2V_4(t_{s2})/\lambda_1(Q)}}{b_1 - 1} \leq ([\sqrt{2(\delta_2/1 + \delta_2)^k V_4(t_{s1})/\lambda_1(Q)}]/[b_1 - 1])$. Note that $\hat{t}_{s2}$ is very small. Then, the entire proof has been completed. ■

C. Fixed-Time Distributed Consensus Tracking Observer Under Directed Communication

Assumption 2: The topology subgraph $\mathcal{G}$ for the group of agents is directed and contains a spanning tree, where the leader is denoted as the root node. Note that, the subgraph $\mathcal{G}_s$ does not need to be strongly connected or contain a spanning tree.

Lemma 3 [49]: Under Assumption 2, $H = L + B$ is of full rank. Furthermore, define $p = [p_1, \dots, p_n]^T = H^{-T} \mathbf{1}_n$, $P = \text{diag}\{p_i\}$, and $Q = ((H^T P + PH)/2)$. Then, P and Q are both positive definite.

In this section, a fixed-time distributed observer under dynamics (6) and (7) with directed communication is designed. Denote the real tracking errors as $\tilde{x}_i = x_i - x_0$ and $\tilde{v}_i = v_i - v_0$. Then, the fixed-time distributed observers α_i and β_i are developed to estimate $\tilde{x}_i$ and $\tilde{v}_i$ in the fixed time. Let $\bar{d} = \max_i \{ \sum_{j=0}^n a_{ij}(d_i - d_j) \}$, $p_{\max} = \max_i \{p_i\}$, $p_{\min} = \min_i \{p_i\}$, $\eta(t) = (p_{\max}/[4\lambda_1(Q)])h(t)$, and $d_0 = 0$. Then, the observer is designed as follows:

$$\begin{aligned} \dot{\alpha}_i &= \beta_i - 2[\eta(t) + 2] \left\{ \sum_{j=0}^n a_{ij} [(\alpha_i - \alpha_j) - (x_i - x_j)] \right\} \\ &\quad - b_1 [\eta(t) + 2] \text{sgn} \left\{ \sum_{j=0}^n a_{ij} [(\alpha_i - \alpha_j) - (x_i - x_j)] \right\} \\ \dot{\beta}_i &= u_i - 2[\eta(t) + 2] \left\{ \sum_{j=0}^n a_{ij} [(\beta_i - \beta_j) - (v_i - v_j)] \right\} \\ &\quad - c_1 [\eta(t) + 2] \text{sgn} \left\{ \sum_{j=0}^n a_{ij} [(\beta_i - \beta_j) - (v_i - v_j)] \right\} \end{aligned} \quad (41)$$

where $\alpha_0 = \beta_0 = 0$, and parameters b_1 and c_1 are positive constants satisfying $b_1 \geq (p_{\max}/[\lambda_1(Q)])$ and $c_1 \geq ([p_{\max}(\bar{d} + u_{\max})]/[\lambda_1(Q)])$.

Theorem 3: With the given dynamics (6) and (7) and observer (41), under Assumption 2, α_i and β_i converge to $\tilde{x}_i$ and $\tilde{v}_i$ within the fixed-time $T_{\max} = t_{s2}$, that is, [see (28)].

Proof: Let $\tilde{\alpha}_i = \alpha_i - \tilde{x}_i$ and $\tilde{\beta}_i = \beta_i - \tilde{v}_i$, and then one has

$$\begin{aligned} \dot{\tilde{\alpha}}_i &= \tilde{\beta}_i - 2[\eta(t) + 2] \sum_{j=0}^n a_{ij} (\tilde{\alpha}_i - \tilde{\alpha}_j) \\ &\quad - b_1 [\eta(t) + 2] \text{sgn} \left[\sum_{j=0}^n a_{ij} (\tilde{\alpha}_i - \tilde{\alpha}_j) \right] \\ \dot{\tilde{\beta}}_i &= -2[\eta(t) + 2] \sum_{j=0}^n a_{ij} (\tilde{\beta}_i - \tilde{\beta}_j) \\ &\quad - c_1 [\eta(t) + 2] \text{sgn} \left[\sum_{j=0}^n a_{ij} (\tilde{\beta}_i - \tilde{\beta}_j) \right] - d_i + u_0. \end{aligned} \quad (42)$$

In order to clarify the main idea clearly and simplify the proof, let $z_i = \sum_{j=0}^n a_{ij} (\tilde{\beta}_i - \tilde{\beta}_j)$, and then $\tilde{\beta}_i$ in (42) can be rewritten as

$$\dot{\tilde{\beta}}_i = -2[\eta(t) + 2] z_i - c_1 [\eta(t) + 2] \text{sgn}(z_i) - d_i + u_0. \quad (43)$$

Differentiate z_i against time, and then we can obtain the following equation:

$$\begin{aligned} \dot{z}_i &= \sum_{j=0}^n a_{ij} (\dot{\tilde{\beta}}_i - \dot{\tilde{\beta}}_j) \\ &= -2[\eta(t) + 2] \sum_{j=0}^n a_{ij} (z_i - z_j) \\ &\quad - c_1 [\eta(t) + 2] \left\{ \sum_{j=0}^n a_{ij} [\text{sgn}(z_i) - \text{sgn}(z_j)] \right\} \\ &\quad - \sum_{j=0}^n a_{ij} (d_i - d_j) + a_{i0} u_0. \end{aligned} \quad (44)$$

Let $z = [z_1, \dots, z_n]^T$ and $\tilde{\beta} = [\tilde{\beta}_1, \dots, \tilde{\beta}_n]^T$, and then one has $z = H\tilde{\beta}$, where H is a nonsingular matrix. Therefore, if vector z converges to zero, vector $\tilde{\beta}$ will converge to zero as well. Then, a Lyapunov candidate function is constructed as follows:

$$V_5 = \sum_{i=1}^n p_i (z_i^2 + c_1 |z_i|). \quad (45)$$

Since P is a positive-definite diagonal matrix, one has each $p_i > 0$. If $V_5 = 0$, one can conclude each $z_i = 0$. Therefore, the Lyapunov candidate function is well constructed. Differentiate V_5 against time, and then one has

$$\begin{aligned} \dot{V}_5 &= \sum_{i=1}^n p_i [2z_i + c_1 \text{sgn}(z_i)] \dot{z}_i \\ &= \sum_{i=1}^n p_i [2z_i + c_1 \text{sgn}(z_i)] \\ &\quad \times \left\{ -2[\eta(t) + 2] \sum_{j=0}^n a_{ij} (z_i - z_j) \right. \\ &\quad \left. - c_1 [\eta(t) + 2] \sum_{j=0}^n a_{ij} [\text{sgn}(z_i) - \text{sgn}(z_j)] \right. \\ &\quad \left. - \sum_{j=0}^n a_{ij} (d_i - d_j) + a_{i0} u_0 \right\}. \end{aligned} \quad (46)$$

In order to give a better understanding of the proof, we separate (46) into two parts, where one part contains the term $-[\eta(t) + 2]$ and the other part not. Then, we rewrite (46) as follows:

$$\begin{aligned} \dot{V}_5 &= -[\eta(t) + 2] \sum_{i=1}^n p_i [2z_i + c_1 \text{sgn}(z_i)] \\ &\quad \times \left\{ 2 \sum_{j=0}^n a_{ij} (z_i - z_j) + c_1 \sum_{j=0}^n a_{ij} [\text{sgn}(z_i) - \text{sgn}(z_j)] \right\} \\ &\quad - \sum_{i=1}^n p_i [2z_i + c_1 \text{sgn}(z_i)] \left[\sum_{j=0}^n a_{ij} (d_i - d_j) - a_{i0} u_0 \right]. \end{aligned} \quad (47)$$

In order to clarify the proof, we rewritten part of (47) into a vector form as follows:

$$\begin{aligned} & \sum_{i=1}^n p_i [2z_i + c_1 \text{sgn}(z_i)] \\ & \times \left\{ 2 \sum_{j=0}^n a_{ij} (z_i - z_j) + c_1 \sum_{j=0}^n a_{ij} [\text{sgn}(z_i) - \text{sgn}(z_j)] \right\} \\ & = [2z + c_1 \text{sgn}(z)]^T PH [2z + c_1 \text{sgn}(z)]. \end{aligned} \quad (48)$$

Then, (47) can be rewritten as

$$\begin{aligned} \dot{V}_5 & = -[\eta(t) + 2][2z + c_1 \text{sgn}(z)]^T PH [2z + c_1 \text{sgn}(z)] \\ & \quad - \sum_{i=1}^n p_i [2z_i + c_1 \text{sgn}(z_i)] \left[\sum_{j=0}^n a_{ij} (d_i - d_j) - a_{i0} u_0 \right]. \end{aligned} \quad (49)$$

Due to

$$\begin{aligned} & [2z + c_1 \text{sgn}(z)]^T PH [2z + c_1 \text{sgn}(z)] \\ & = [2z + c_1 \text{sgn}(z)]^T H^T P [2z + c_1 \text{sgn}(z)] \end{aligned} \quad (50)$$

one can conclude

$$\begin{aligned} & [2z + c_1 \text{sgn}(z)]^T PH [2z + c_1 \text{sgn}(z)] \\ & = [2z + c_1 \text{sgn}(z)]^T \frac{PH + H^T P}{2} [2z + c_1 \text{sgn}(z)] \\ & = [2z + c_1 \text{sgn}(z)]^T Q [2z + c_1 \text{sgn}(z)]. \end{aligned} \quad (51)$$

Then, (49) can be rewritten as follows:

$$\begin{aligned} \dot{V}_5 & = -[\eta(t) + 2][2z + c_1 \text{sgn}(z)]^T Q [2z + c_1 \text{sgn}(z)] \\ & \quad - \sum_{i=1}^n p_i [2z_i + c_1 \text{sgn}(z_i)] \left[\sum_{j=0}^n a_{ij} (d_i - d_j) - a_{i0} u_0 \right] \\ & \leq -\lambda_1(Q) [\eta(t) + 2][2z + c_1 \text{sgn}(z)]^T [2z + c_1 \text{sgn}(z)] \\ & \quad - \sum_{i=1}^n p_i [2z_i + c_1 \text{sgn}(z_i)] \left[\sum_{j=0}^n a_{ij} (d_i - d_j) - a_{i0} u_0 \right]. \end{aligned} \quad (52)$$

Due to $[2z + c_1 \text{sgn}(z)]^T [2z + c_1 \text{sgn}(z)] = \sum_{i=1}^n (4z_i^2 + 4c_1 |z_i| + c_1^2)$, $\bar{d} = |\max_i \{\sum_{j=0}^n a_{ij} (d_i - d_j)\}|$, $p_{\max} = \max_i \{p_i\}$, and $|u_0| \leq u_{\max}$, one can obtain

$$\begin{aligned} \dot{V}_5 & \leq -\lambda_1(Q) [\eta(t) + 1] \sum_{i=1}^n (4z_i^2 + 4c_1 |z_i| + c_1^2) \\ & \quad - \lambda_1(Q) \sum_{i=1}^n (4z_i^2 + 4c_1 |z_i| + c_1^2) \\ & \quad + (\bar{d} + u_{\max}) p_{\max} \sum_{i=1}^n (2|z_i| + c_1) \\ & \leq -4\lambda_1(Q) [\eta(t) + 1] \sum_{i=1}^n (z_i^2 + c_1 |z_i|) \\ & \quad - \lambda_1(Q) c_1 \sum_{i=1}^n (4|z_i| + c_1) \\ & \quad + (\bar{d} + u_{\max}) p_{\max} \sum_{i=1}^n (2|z_i| + c_1). \end{aligned} \quad (53)$$

Then, according to the precondition $c_1 \geq ([(\bar{d} + u_{\max}) p_{\max}] / [\lambda_1(Q)])$, one can conclude

$$\dot{V}_5 \leq -4\lambda_1(Q) [\eta(t) + 1] \sum_{i=1}^n (z_i^2 + c_1 |z_i|). \quad (54)$$

By introducing the constant $p_{\max}$ into (54), one has

$$\begin{aligned} \dot{V}_5 & \leq -\frac{4\lambda_1(Q)}{p_{\max}} [\eta(t) + 1] \sum_{i=1}^n p_{\max} (z_i^2 + c_1 |z_i|) \\ & \leq -\frac{4\lambda_1(Q)}{p_{\max}} [\eta(t) + 1] \sum_{i=1}^n p_i (z_i^2 + c_1 |z_i|) \\ & = -\frac{4\lambda_1(Q)}{p_{\max}} [\eta(t) + 1] V_5 \\ & = -\frac{4\lambda_1(Q)}{p_{\max}} \eta(t) V_5 - \frac{4\lambda_1(Q)}{p_{\max}} V_5 \\ & \leq -\frac{4\lambda_1(Q)}{p_{\max}} \eta(t) V_5 \\ & = -h(t) V_5. \end{aligned} \quad (55)$$

When $t \in [0, t_{s1})$, $h(t) = h_1(t)$. According to (1) one has

$$\begin{aligned} \lim_{t \rightarrow t_{s1}} V_5 & = \lim_{t \rightarrow t_{s1}} \sum_{i=1}^n p_i (z_i^2 + c_1 |z_i|) \\ & = \lim_{t \rightarrow t_{s1}} \sum_{i=1}^n p_i z_i^2 + \lim_{t \rightarrow t_{s1}} \sum_{i=1}^n c_1 p_i |z_i| \\ & \leq \left(\frac{\delta_1}{1 + \delta_1} \right)^k V_5(0). \end{aligned} \quad (56)$$

Then, one has

$$\lim_{t \rightarrow t_{s1}} \sum_{i=1}^n p_i z_i^2 \leq \left(\frac{\delta_1}{1 + \delta_1} \right)^k V_5(0) \quad (57)$$

and

$$\lim_{t \rightarrow t_{s1}} \sum_{i=1}^n z_i^2 \leq \frac{1}{p_{\min}} \left(\frac{\delta_1}{1 + \delta_1} \right)^k V_5(0). \quad (58)$$

Due to

$$\sum_{i=1}^n z_i^2 = z^T z = \tilde{\beta}^T H^T H \tilde{\beta} \quad (59)$$

and

$$\tilde{\beta}^T H^T H \tilde{\beta} \geq \lambda_1(H^T H) \tilde{\beta}^T \tilde{\beta} = \sum_{i=1}^n \lambda_1(H^T H) \tilde{\beta}_i^2 \quad (60)$$

one can conclude

$$\sum_{i=1}^n \lambda_1(H^T H) \tilde{\beta}_i^2 \leq \sum_{i=1}^n z_i^2 \leq \frac{1}{p_{\min}} \left(\frac{\delta_1}{1 + \delta_1} \right)^k V_5(0). \quad (61)$$

According the above proof, one can make a conclusion that $\lim_{t \rightarrow t_{s1}} \sum_{i=1}^n \tilde{\beta}_i^2 \leq (1/[\lambda_1(H^T H) p_{\min}]) (\delta_1/1 + \delta_1)^k V_5(0)$, and $\lim_{t \rightarrow t_{s1}} |\tilde{\beta}| \leq \sqrt{(1/[\lambda_1(H^T H) p_{\min}]) (\delta_1/1 + \delta_1)^k V_5(0)}$. Therefore, velocity observer β_i successfully obtains the relative error $v_i - v_0$ in the fixed time t_{s1} . When $t \geq t_{s1}$, due to $\dot{V}_5 \leq -([4\lambda_1(Q)]/p_{\max}) V_5$. The Lyapunov candidate function V_5 will continue to converge to zero exponentially. The proof

for velocity observer β_i has been completed. Although V_5 does not perfectly converge to zero at t_{s1} , the stability of position observer α_i is not affected. The proof for position observer α_i is given as follows.

Let $w_i = \sum_{j=0}^n a_{ij}(\tilde{\alpha}_i - \tilde{\alpha}_j)$. Substitute w_i into (42), and then one obtains

$$\dot{\tilde{\alpha}}_i = \tilde{\beta}_i - 2[\eta(t) + 2]w_i - b_1[\eta(t) + 2]\text{sgn}(w_i). \quad (62)$$

Differentiate w_i against time, and then one has

$$\begin{aligned} \dot{w}_i &= z_i - 2[\eta(t) + 2] \sum_{j=0}^n a_{ij}(w_i - w_j) \\ &\quad - b_1[\eta(t) + 2] \left\{ \sum_{j=0}^n a_{ij}[\text{sgn}(w_i) - \text{sgn}(w_j)] \right\}. \end{aligned} \quad (63)$$

Construct a Lyapunov candidate as follows:

$$V_6 = \sum_{i=1}^n p_i (w_i^2 + b_1 |w_i|). \quad (64)$$

Let $w = [w_1, \dots, w_n]^T$ and $\tilde{\alpha} = [\tilde{\alpha}_1, \dots, \tilde{\alpha}_n]^T$, and then one has $w = H\tilde{\alpha}$. Due to the full rank of matrix H , if vector w converges to zero, vector $\tilde{\alpha}$ converge to zero as well. Moreover, if V_6 converges to zero, vector w converges to zero as well. Therefore, the Lyapunov candidate function is well constructed. Differentiate V_6 against time, and then one has

$$\dot{V}_6 = \sum_{i=1}^n p_i [2w_i + b_1 \text{sgn}(w_i)] \dot{w}_i. \quad (65)$$

Substitute (63) into (65), and then one has

$$\begin{aligned} \dot{V}_6 &= -[\eta(t) + 2] \sum_{i=1}^n p_i [2w_i + b_1 \text{sgn}(w_i)] \\ &\quad \times \left\{ 2 \sum_{j=0}^n a_{ij}(w_i - w_j) + b_1 \sum_{j=0}^n a_{ij}[\text{sgn}(w_i) - \text{sgn}(w_j)] \right\} \\ &\quad - \sum_{i=1}^n p_i z_i [2w_i + b_1 \text{sgn}(w_i)]. \end{aligned} \quad (66)$$

According to (48) and (51), one has

$$\begin{aligned} \dot{V}_6 &= -[\eta(t) + 2] [2w + b_1 \text{sgn}(w)]^T Q [2w + b_1 \text{sgn}(w)] \\ &\quad - \sum_{i=1}^n p_i z_i [2w_i + b_1 \text{sgn}(w_i)] \\ &\leq -\lambda_1(Q) [\eta(t) + 2] [2w + b_1 \text{sgn}(w)]^T [2w + b_1 \text{sgn}(w)] \\ &\quad - \sum_{i=1}^n p_i z_i [2w_i + b_1 \text{sgn}(w_i)] \\ &= -\lambda_1(Q) [\eta(t) + 2] \sum_{i=1}^n (4w_i^2 + 4b_1 |w_i| + b_1^2) \\ &\quad - \sum_{i=1}^n p_i z_i [2w_i + b_1 \text{sgn}(w_i)] \\ &\leq -4\lambda_1(Q) [\eta(t) + 1] \sum_{i=1}^n (w_i^2 + b_1 |w_i|) \end{aligned}$$

$$\begin{aligned} &- b_1 \lambda_1(Q) \sum_{i=1}^n (2|w_i| + b_1) \\ &- \sum_{i=1}^n p_i z_i [2w_i + b_1 \text{sgn}(w_i)]. \end{aligned} \quad (67)$$

When $t \geq t_{s1}$, according to (58), one has $|z_i| \ll 1$. Then, one can conclude

$$\begin{aligned} \dot{V}_6 &\leq -4\lambda_1(Q) [\eta(t) + 1] \sum_{i=1}^n (w_i^2 + b_1 |w_i|) \\ &\quad - b_1 \lambda_1(Q) \sum_{i=1}^n (2|w_i| + b_1) \\ &\quad + p_{\max} \sum_{i=1}^n [2|w_i| + b_1]. \end{aligned} \quad (68)$$

Due to the precondition $b_1 \geq (p_{\max}/[\lambda_1(Q)])$, one can obtain

$$\begin{aligned} \dot{V}_6 &\leq -4\lambda_1(Q) [\eta(t) + 1] \sum_{i=1}^n (w_i^2 + b_1 |w_i|) \\ &\leq -h(t) V_6 - \frac{4\lambda_1(Q)}{p_{\max}} V_6 \\ &\leq -h(t) V_6. \end{aligned} \quad (69)$$

Then, the following proof is the same as (55), and one can obtain $\lim_{t \rightarrow t_{s2}} |\tilde{\alpha}_i| \leq \sqrt{(1/[\lambda_1(H^T H) p_{\min}]) (\delta_2/1 + \delta_2)^k V_6(t_{s1})}$. When $t \geq t_{s2}$, due to $\dot{V}_6 \leq -(4\lambda_1(Q)/p_{\max}) V_6$, the Lyapunov candidate function V_6 will converge to zero exponentially. Therefore, the proof of Theorem 3 has been finished. ■

D. Fixed-Time Distributed Averaging Tracking Observer

Assumption 3: The topology graph for the n agents is undirected and connected. Each agent can only receive information from the corresponding reference signal.

Assumption 4: It is assumed that the initial states of the proposed observer satisfy $\sum_{i=1}^n \alpha_i(0) = \sum_{i=1}^n r_i(0)$, and $\sum_{i=1}^n \beta_i(0) = \sum_{i=1}^n f_i(0)$, where $f_i(0)$ and $r_i(0)$ are the initial velocity and position of the reference signals, respectively.

Lemma 4 [48]: If $L \in \mathbb{R}^{n \times n}$ is a Laplacian matrix of a connected undirected graph and $D \in \mathbb{R}^{n \times m}$ is the corresponding relative incidence matrix. Then, for any vector $z \in \mathbb{R}^n$, one has

$$z^T L D \text{sgn}(D^T z) \geq \lambda_2(L) z^T D \text{sgn}(D^T z). \quad (70)$$

In this section, the fixed-time distributed observers α_i and β_i are designed to estimate the average position $\bar{r}$ and average velocity $\bar{f}$ of multiple signals within the fixed time independent of the initial system states, respectively. The specific formulation of the distributed observer is given as follows:

$$\begin{aligned} \dot{\alpha}_i &= -\eta(t) \sum_{j=1}^n a_{ij}(\alpha_i - \alpha_j) \\ &\quad - b_1 \sum_{j=1}^n a_{ij} \text{sgn}(\alpha_i - \alpha_j) + \beta_i \end{aligned}$$

$$\begin{aligned}\dot{\beta}_i &= -\eta(t) \sum_{j=1}^n a_{ij}(\beta_i - \beta_j) \\ &\quad - c_1 \sum_{j=1}^n a_{ij} \text{sgn}(\beta_i - \beta_j) + a_i^r\end{aligned}\quad (71)$$

where $\eta(t) = (1/[2\lambda_2(L)])h(t)$, and parameters b_1 and c_1 are positive constants satisfying $b_1 \geq (1/[\lambda_2(L)])$, $c_1 > (2a_{\max}/[\lambda_2(L)])$.

Theorem 4: With the given dynamics (7) and (9) and observer (71), under Assumptions 3 and 4, observers α_i and β_i converge to $\bar{r}$ and $\bar{f}$ within a fixed-time $T_{\max} = t_{s2}$, that is

$$\begin{cases} \lim_{t \rightarrow t_{s1}} |\beta_i - \bar{f}| \leq o_1 \\ \lim_{t \rightarrow t_{s2}} |\alpha_i - \bar{r}| \leq o_2 \\ \lim_{t \rightarrow \infty} |\beta_i - \bar{f}| = 0 \\ \lim_{t \rightarrow \infty} |\alpha_i - \bar{r}| = 0 \end{cases}\quad (72)$$

where the parameters o_1 and o_2 are small positive constants close to zero and can be limited to a desired level.

Proof: Since the communication topology graph is undirected, one has $a_{ij} = a_{ji}$. According to (71), one can obtain

$$\begin{cases} \sum_{i=1}^n \dot{\alpha}_i = \sum_{i=1}^n \beta_i \\ \sum_{i=1}^n \dot{\beta}_i = \sum_{i=1}^n a_i^r. \end{cases}\quad (73)$$

Due to $\sum_{i=1}^n \dot{\beta}_i = \sum_{i=1}^n a_i^r$ and $\sum_{i=1}^n \beta_i(0) = \sum_{i=1}^n f_i(0)$, one can obtain $\sum_{i=1}^n \beta_i = \sum_{i=1}^n f_i$. Moreover, due to $\sum_{i=1}^n \dot{\alpha}_i = \sum_{i=1}^n \beta_i = \sum_{i=1}^n f_i$ and $\sum_{i=1}^n \alpha_i(0) = \sum_{i=1}^n r_i(0)$, one can conclude $\sum_{i=1}^n \alpha_i = \sum_{i=1}^n r_i$. Therefore, if observer α_i and observer β_i achieve consensus in the fixed time t_{s2} , they successfully observe the average position and velocity of reference signals.

We written (71) in the vector form as follows:

$$\begin{aligned}\dot{\alpha} &= -\eta(t)L\alpha - b_1 D \text{sgn}(D^T \alpha) + \beta \\ \dot{\beta} &= -\eta(t)L\beta - c_1 D \text{sgn}(D^T \beta) + a^r\end{aligned}\quad (74)$$

where $a^r = [a_1^r, \dots, a_n^r]^T$. Then, we construct a Lyapunov candidate as $V_7 = (1/2)\beta^T L \beta$. Note that $(1/2)\beta^T L \beta = \sum_{i=1}^n \sum_{j=1}^n (\beta_i - \beta_j)^2$. Differentiate V_7 against time and then one has

$$\begin{aligned}\dot{V}_7 &= \beta^T L \dot{\beta} \\ &= \beta^T L [-\eta(t)L\beta - c_1 D \text{sgn}(D^T \beta) + a^r] \\ &\leq -\eta(t)\lambda_2(L)\beta^T L \beta - c_1 \beta^T L D \text{sgn}(D^T \beta) \\ &\quad + \beta^T L a^r.\end{aligned}\quad (75)$$

According to Lemma 4 and the precondition $c_1 > (2a_{\max}/[\lambda_2(L)])$, one has

$$\begin{aligned}\dot{V}_7 &\leq -\eta(t)\lambda_2(L)\beta^T L \beta - c_1 \lambda_2(L)\beta^T D \text{sgn}(D^T \beta) + \beta^T L a^r \\ &= -h(t)V_7 - c_1 \lambda_2(L)\|D^T \beta\|_1 + (D^T \beta)^T D^T a^r \\ &\leq -h(t)V_7 - (c_1 \lambda_2(L) - 2a_{\max})\|D^T \beta\|_1 \\ &\leq -h(t)V_7.\end{aligned}\quad (76)$$

When $t \in [0, t_{s1})$, $h(t) = h_1(t)$. According to (1), one has $\lim_{t \rightarrow t_{s1}} V_7 = (1/2) \sum_{i=1}^n \sum_{j=1}^n a_{ij}(\beta_i - \beta_j)^2 \leq$

$(\delta_1/1 + \delta_1)^k V_7(0)$. Then, one can conclude

$$\begin{aligned}\lim_{t \rightarrow t_{s1}} \left| \beta_i - \frac{1}{n} \sum_{i=1}^n f_i \right| &= \lim_{t \rightarrow t_{s1}} \left| \beta_i - \frac{1}{n} \sum_{j=1}^n \beta_j \right| \\ &\leq \lim_{t \rightarrow t_{s1}} \max\{|\beta_i - \beta_j| \mid \forall i, j \in n\} \\ &\leq \sqrt{2 \left(\frac{\delta_1}{1 + \delta_1} \right)^k V_7(0)}.\end{aligned}\quad (77)$$

When $t \geq t_{s1}$, $\dot{V}_7 \leq -(c_1 \lambda_2(L) - 2a_{\max})\|D^T \beta\|_1 \leq -(c_1 \lambda_2(L) - 2a_{\max})\sqrt{2\lambda_2(L)V_7}$. Then, one has V_7 will converge quickly to zero in the finite time after t_{s1} [see the proof of (36)].

Construct the Lyapunov candidate as $V_8 = (1/2)\alpha^T L \alpha$ and differentiate it against time

$$\begin{aligned}\dot{V}_8 &= \alpha^T L \dot{\alpha} \\ &= \alpha^T L [-\eta(t)L\alpha - b_1 D \text{sgn}(D^T \alpha) + \beta] \\ &= -\eta(t)\alpha^T L L \alpha - b_1 \alpha^T L D \text{sgn}(D^T \alpha) + \alpha^T L \beta \\ &\leq -\lambda_2(L)\eta(t)\alpha^T L \alpha - b_1 \|D^T \alpha\|_1 + (D^T \alpha)^T D^T \beta \\ &\leq -h(t)V_8\end{aligned}\quad (78)$$

where the second inequality holds due to $|\beta_i - \beta_j| \ll 1$ and the precondition $b_1 \geq 1$. The following proof is the same as that of (76) and hence omitted. Then, the proof of Theorem 4 has been finished. ■

Remark 4: All the observers proposed in this article can be extended to that for high-order multiagent systems by using more time-based generators and more integrators.

E. Distributed Consensus Tracking and Distributed Average Tracking Control

After designing the observers, we will show how to design the fixed-time controllers by using the information provided by the observers. Two new time-based generators $\xi_3(t)$ and $\xi_4(t)$ are designed as follows:

$$\xi(t) = \begin{cases} \xi_3(t), & t_b \leq t < t_{s3} \\ \xi_4(t), & t_{s3} \leq t < t_{s4} \\ 1, & t_{s4} \leq t < \infty \end{cases}\quad (79)$$

where $t_b \geq t_{s2}$. According to (2), the corresponding $h(t)$ is constructed as follows:

$$h(t) = \begin{cases} h_3(t) = k \frac{\dot{\xi}_3(t)}{1 - \xi_3(t) + \delta_3}, & t_b \leq t < t_{s3} \\ h_4(t) = k \frac{\dot{\xi}_4(t)}{1 - \xi_4(t) + \delta_4}, & t_{s3} \leq t < t_{s4} \\ 0, & t_{s4} \leq t < \infty. \end{cases}\quad (80)$$

First, the detail of the controller for the distributed consensus tracking problem is shown follows.

Theorem 5: Under dynamics (6) and (7) and Assumption 1 (Assumption 2), with given observer (27) [observer (41)] and controller

$$u_i = \begin{cases} = 0, & t \in [0, t_b) \\ = -\frac{1}{2}\dot{h}(t)\alpha_i - \left(\frac{1}{2}h(t) + 1\right)\beta_i \\ = -\frac{1}{2}h(t)s_i - \rho \text{sgn}(s_i), & t \geq t_b \end{cases}\quad (81)$$

where s_i is the sliding surface, and $s_i = ((1/2)h(t) + 1)\alpha_i + \beta_i$, $\rho \geq d_{\max} + u_{\max} + 1$; the fixed-time distributed consensus tracking problem for double-integrator-type multiagent systems is solved. Furthermore, the upper bounded convergence time is $T_{\max} = t_{s4}$.

Proof: When $t \geq t_b$, we consider $\alpha_i = x_i - x_0$ and $\beta_i = v_i - v_0$. Differentiate s_i against time and then one has

$$\begin{aligned}\dot{s}_i &= \frac{1}{2}\dot{h}(t)\alpha_i + \left(\frac{1}{2}h(t) + 1\right)\dot{\alpha}_i + \dot{\beta}_i \\ &= \frac{1}{2}\dot{h}(t)\alpha_i + \left(\frac{1}{2}h(t) + 1\right)\beta_i + \dot{v}_i - \dot{v}_0 \\ &= \frac{1}{2}\dot{h}(t)\alpha_i + \left(\frac{1}{2}h(t) + 1\right)\beta_i + u_i + d_i - u_0.\end{aligned}\quad (82)$$

Substitute controller (81) into (82), and then one has

$$\dot{s}_i = -\frac{1}{2}h(t)s_i - \rho \operatorname{sgn}(s_i) + d_i - u_0. \quad (83)$$

Choose $V_1 = (1/2)s_i^2$ as the Lyapunov candidate functions. Differentiate V_1 against time and then one has

$$\begin{aligned}\dot{V}_1 &= s_i \dot{s}_i \\ &= s_i \left[-\frac{1}{2}h(t)s_i - \rho \operatorname{sgn}(s_i) + d_i - u_0 \right] \\ &= -h(t)V_1 - \rho|s_i| + (d_i - u_0)s_i \\ &\leq -h(t)V_1 - (\rho - d_{\max} - u_{\max})|s_i|.\end{aligned}\quad (84)$$

Due to precondition $\rho \geq d_{\max} + u_{\max} + 1$, one has

$$\begin{aligned}\dot{V}_1 &\leq -h(t)V_1 - |s_i| \\ &\leq -h(t)V_1.\end{aligned}\quad (85)$$

According to the property of (1), one can come to the conclusion that $\lim_{t \rightarrow t_{s3}} V_1 \leq (\delta_3/1 + \delta_3)^k V_1(t_b)$ and $\lim_{t \rightarrow t_{s3}} |s_i| \leq \sqrt{2(\delta_3/1 + \delta_3)^k V_1(t_b)}$. Let $z_1 = \alpha_i$ and $z_2 = \beta_i$, and construct the Lyapunov candidate function as $V_2 = (1/2)z_i^2$. Then, the following proof is the same as Theorem 1 and omitted [see the proof of (20)]. ■

Next, the detail of the controller for the fixed-time distributed average tracking problem is shown as follows.

Theorem 6: Under dynamics (7) and (9) and Assumption 3, with given observer (71) and controller

$$u_i = \begin{cases} = 0, & t \in [0, t_b) \\ = -\frac{1}{2}\dot{h}(t)(x_i - \alpha_i) - \left(\frac{1}{2}h(t) + 1\right)(v_i - \beta_i) \\ \quad - \frac{1}{2}h(t)s_i - \rho \operatorname{sgn}(s_i), & t \geq t_b \end{cases} \quad (86)$$

where s_i is the sliding-mode surface, and $s_i = ((1/2)h(t) + 1)(x_i - \alpha_i) + (v_i - \beta_i)$, $\rho \geq d_{\max} + a_{\max} + 1$; the fixed-time distributed average tracking problem for double-integrator-type multiagent systems is solved. Furthermore, the upper bounded convergence time is $T_{\max} = t_{s4}$.

Proof: When $t \geq t_b$, we consider $\alpha_i = \bar{r}$ and $\beta_i = \bar{f}$. Differentiate s_i against time and then one has

$$\begin{aligned}\dot{s}_i &= \frac{1}{2}\dot{h}(t)(x_i - \alpha_i) + \left(\frac{1}{2}h(t) + 1\right)(v_i - \beta_i) \\ &\quad + u_i + d_i - \bar{a}.\end{aligned}\quad (87)$$

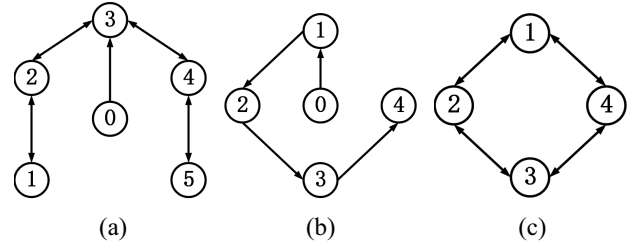


Fig. 1. Topology graph for multiagent systems. (a) Topology graph 1. (b) Topology graph 2. (c) Topology graph 3.

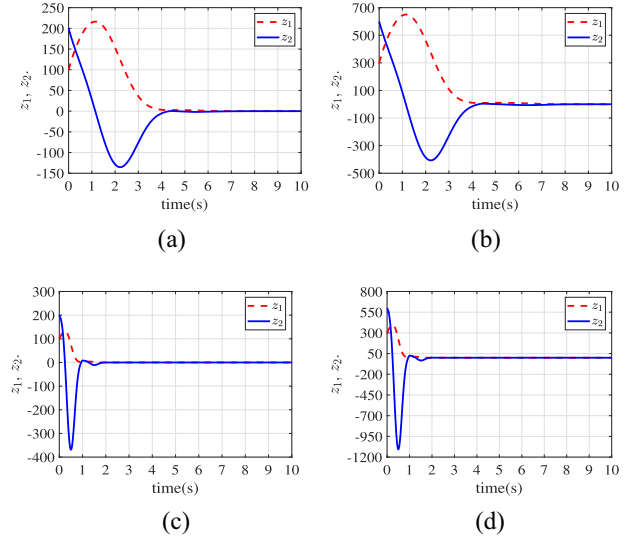


Fig. 2. Fixed-time sliding-mode control. (a) Comparison group $c_1(a)$. (b) Comparison group $c_1(b)$. (c) Comparison group $c_2(a)$. (d) Comparison group $c_2(b)$.

Substitute controller (86) into (87), and then one has

$$\dot{s}_i = -\frac{1}{2}h(t)s_i - \rho \operatorname{sgn}(s_i) + d_i - \bar{a}. \quad (88)$$

The other part of the proof is the same as Theorem 5 and hence omitted [see the proof of (83)]. Then, the entire proof has been completed. ■

IV. NUMERICAL SIMULATIONS

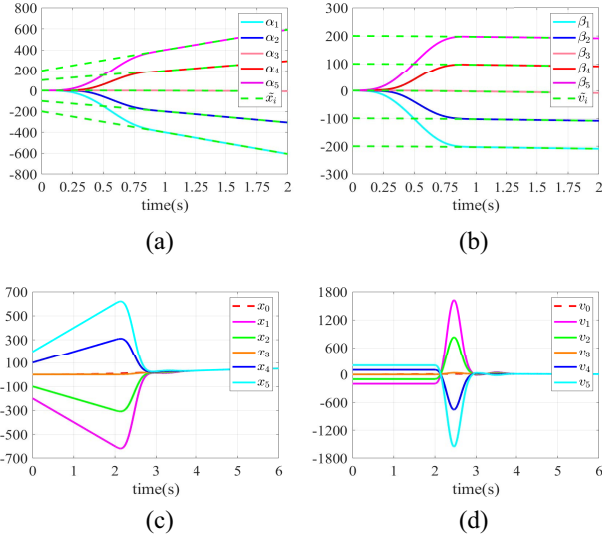
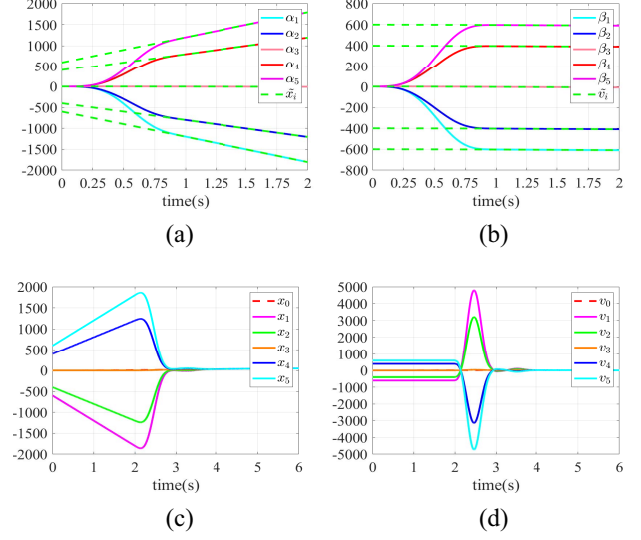
Three kinds of communication topology graphs for multiagent systems are shown in Fig. 1, which will be used in the following experiments.

Example 1: A simulation for Theorem 1 is given here and the results are shown in Fig. 2. The experiment is divided into two comparison groups (c_1 and c_2) and the parameter settings are shown in Table I. The initial states in $c_1(a)$ and $c_2(a)$ are $z_1(0) = 100$ and $z_2(0) = 200$, and those in $c_1(b)$ and $c_2(b)$ are $z_1(0) = 300$ and $z_2(0) = 600$, and disturbance $\varrho = \sin(t)$. By comparing the results between $c_1(a)$ and $c_1(b)$ or $c_2(a)$ and $c_2(b)$ in Fig. 2, we conclude the upper bounded convergence time is not influenced by the initial states. Comparing the results between $c_1(a)$ and $c_2(a)$ or $c_1(b)$ and $c_2(b)$ in Fig. 2, we conclude that the upper bounded convergence time $T_{\max}$ can be easily adjusted by modifying the parameter t_{s2} .

Example 2: A simulation for Theorems 2 and 5 under undirected communication is shown in Figs. 3 and 4. Consider

TABLE I
PARAMETER SETTINGS

Experiment	group	k	δ_1	δ_2	δ_3	δ_4	ρ	t_{s1}	t_{s2}	t_b	t_{s3}	t_{s4}	b_1	c_1
Example 1	c_1	2	0.1	0.1	-	-	2	4.5s	9s	-	-	-	-	-
	c_2	2	0.1	0.1	-	-	2	1s	2s	-	-	-	-	-
Example 2	c_1/c_2	2	0.1	0.1	0.01	0.01	8	1s	2s	2s	3s	4s	1	8
Example 3	c_1/c_2	1.2	0.01	0.01	0.01	0.01	22	3s	6s	6s	9s	12s	7	134
Example 4	c_1/c_2	2	0.1	0.1	0.01	0.01	62	1s	2s	2s	4s	6s	1	123

Fig. 3. Experimental results of fixed-time distributed consensus tracking under undirected communication (comparison group c_1). (a) Position observer. (b) Velocity observer. (c) Position controller. (d) Velocity controller.Fig. 4. Experimental results of fixed-time distributed consensus tracking under undirected communication (comparison group c_2). (a) Position observer. (b) Velocity observer. (c) Position controller. (d) Velocity controller.

a multiagent system described by (6) and (7) with the communication topology depicted in the first subgraph of Fig. 1. The simulation contains two comparison groups (c_1 and c_2), and the parameter settings are shown in Table I. In c_1 , we set leader control input $u_0 = 1 + 5\sin(t)$, disturbance $d_i = \sin(t)$, and system initial states $x_0(0) = v_0(0) = 0$, $x_1(0) = v_1(0) = -200$, $x_2(0) = v_2(0) = -100$, $x_3(0) = v_3(0) = 0$, $x_4(0) = v_4(0) = 100$, $x_5(0) = v_5(0) = 200$. In c_2 , we reset the initial states as $x_1(0) = v_1(0) = -600$, $x_2(0) = v_2(0) = -400$, $x_3(0) = v_3(0) = 0$, $x_4(0) = v_4(0) = 400$, and $x_5(0) = v_5(0) = 600$. By comparing the results of the two comparison groups in Figs. 3 and 4, we conclude the observer successfully observe the object value in t_{s2} and the controller drive the multiagent system to achieve consensus with the leader in $T_{\max} = t_{s4}$. Moreover, the upper bounded convergence time is not influenced by system initial states.

Example 3: A simulation for Theorems 3 and 5 under directed communication topology is shown in Figs. 5 and 6. Consider a multiagent system described by (6) and (7) with the communication topology depicted in the second subgraph of Fig. 1. The experiment contains two comparison groups (c_1 and c_2), and the parameter settings are shown in Table I. In c_1 , we set leader control input $u_0 = 18\sin(10t) + 2$, disturbance $d_i = \sin(t)$, and system initial states $x_0(0) = v_0(0) = 300$, $x_1(0) = v_1(0) = -100$, $x_2(0) = v_2(0) = -50$, $x_3(0) = v_3(0) = 0$, $x_4(0) = v_4(0) = 50$, and $x_5(0) = v_5(0) = 100$. In c_2 , we reset the initial states as $x_1(0) = v_1(0) = -500$,

$x_2(0) = v_2(0) = -400$, $x_3(0) = v_3(0) = 0$, $x_4(0) = v_4(0) = 400$, and $x_5(0) = v_5(0) = 500$. By comparing the results of two comparison groups in Figs. 5 and 6, we conclude the observer successfully observe the object value and the controller drive the multiagent system to achieve consensus with the leader in the fixed time. Moreover, the upper bounded convergence time is not influenced by system initial states.

Example 4: A simulation for Theorems 4 and 6 is given in Figs. 7 and 8. Consider a multiagent system described by (7) and (9) with the communication topology depicted in the third subgraph of Fig. 1. The experiment contains two comparison groups (c_1 and c_2). In c_1 , we set the acceleration of the i th reference signal as $a_i^r = 20\sin(5t) + 10i$, the reference signal initial states as $r_1(0) = f_1(0) = 200$, $r_2(0) = f_2(0) = 100$, $r_3(0) = f_3(0) = -100$, and $r_4(0) = f_4(0) = -200$, and the system initial states as $x_1(0) = v_1(0) = -100$, $x_2(0) = v_2(0) = -200$, $x_3(0) = v_3(0) = 100$, and $x_4(0) = v_4(0) = 300$. In c_2 , we reset the reference signal initial states as $r_1(0) = f_1(0) = 600$, $r_2(0) = f_2(0) = 400$, $r_3(0) = f_3(0) = -400$, and $r_4(0) = f_4(0) = -600$, the system initial states $x_1(0) = v_1(0) = 600$, $x_2(0) = v_2(0) = 500$, $x_3(0) = v_3(0) = 400$, and $x_4(0) = v_4(0) = 300$. By comparing the results of the two comparison groups in Figs. 7 and 8, we conclude the observers successfully observe the mean value of multiple reference signals in t_{s2} and the controllers drive the multiagent system to achieve consensus with the mean value of multiple reference signals in $T_{\max} = t_{s4}$. Moreover,

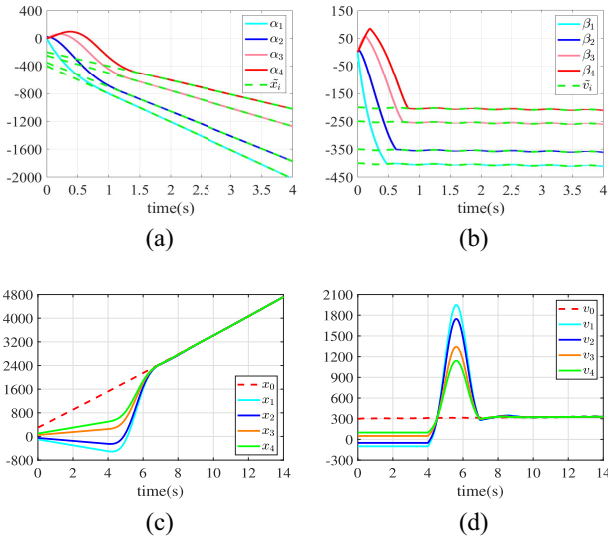


Fig. 5. Experimental results of fixed-time distributed consensus tracking under directed communication (comparison group c_1). (a) Position observer. (b) Velocity observer. (c) Position controller. (d) Velocity controller.

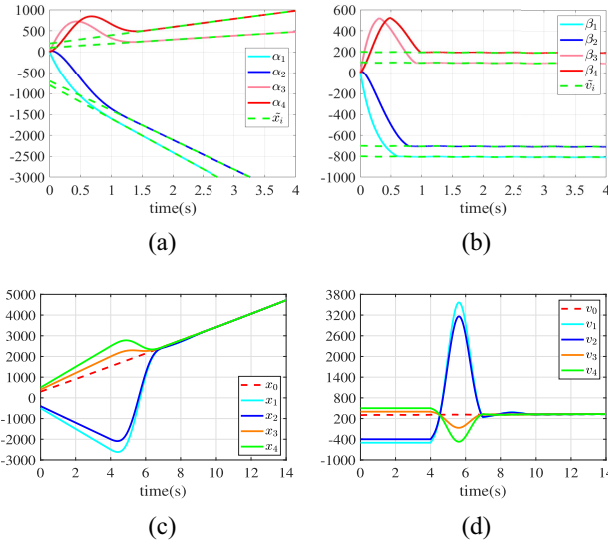


Fig. 6. Experimental results of fixed-time distributed consensus tracking under directed communication (comparison group c_2). (a) Position observer. (b) Velocity observer. (c) Position controller. (d) Velocity controller.

the upper bounded convergence time is not changed with the variation system initial states.

Remark 5: Increasing the value of parameter k or decreasing that of parameter δ can improve the convergence accuracy at the predefined convergence time, but it will also increase the accuracy demand of the control system. The predefined convergence time t_s can be designed arbitrarily, but a small t_s has to demand a large control input. The main effect of other parameters is to guarantee the proof correct, and have a limited influence on the real convergence time.

V. CONCLUSION

In this article, both the fixed-time distributed consensus tracking and the fixed-time distributed average tracking problems for double-integrator-type multiagent systems were

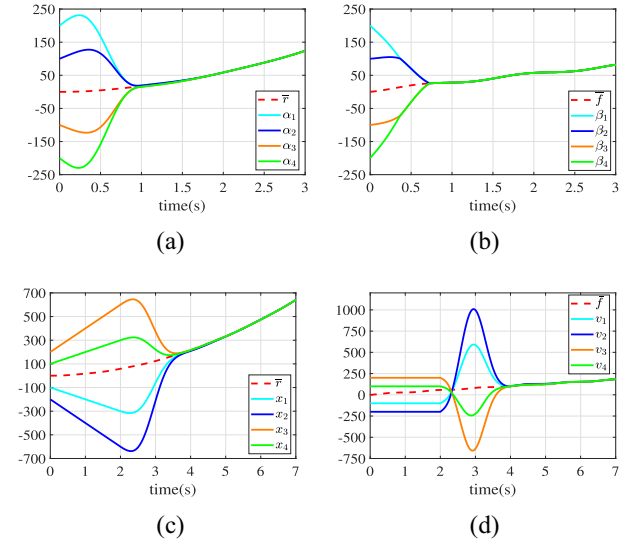


Fig. 7. Experiment results of fixed-time distributed average tracking (comparison group c_1). (a) Position observer. (b) Velocity observer. (c) Position controller. (d) Velocity controller.

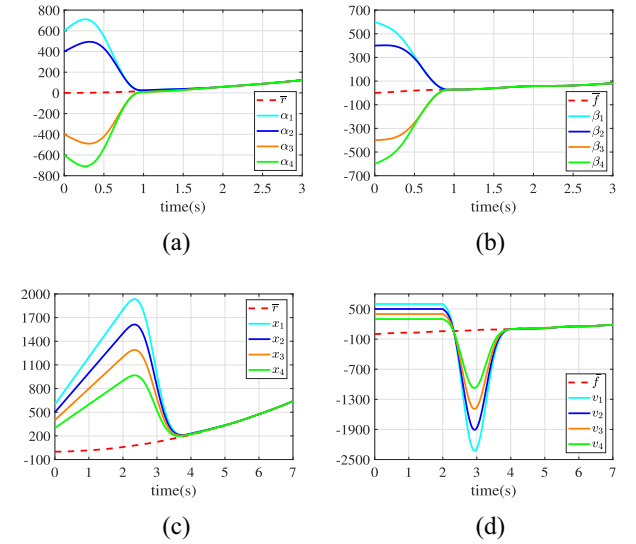


Fig. 8. Experiment results of fixed-time distributed average tracking (comparison group c_2). (a) Position observer. (b) Velocity observer. (c) Position controller. (d) Velocity controller.

solved by using time-based generators. Different from traditional fixed-time methods, the time-based generator approach can avoid control input saturation and determine the fixed time without complicated computation. In the future, we try to extend the proposed fixed-time sliding-mode control method to Euler–Lagrange systems. By combining the sliding-mode control method of Euler–Lagrange systems and the proposed observers, the fixed-time distributed consensus tracking and distributed average tracking problems for multiple Euler–Lagrange systems can be solved.

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